Guideline for the Assembling Procedure

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This report is a guideline for the assembling procedure of the flexible aircraft flight dynamic model.

From the lecture, we know that the linearized flight dynamic and kinematic equations are:

$$\boldsymbol{M}_{rr}\dot{\boldsymbol{x}}_{rr} = \boldsymbol{A}_{rr}\boldsymbol{x}_{rr} + \boldsymbol{B}_{rr}\boldsymbol{\delta}_T + \begin{pmatrix} \boldsymbol{0} \\ \Delta\boldsymbol{F}_y \\ \boldsymbol{0} \\ \Delta\boldsymbol{M} \end{pmatrix}_{\text{wing}} \tag{1}$$

where $\boldsymbol{x}_{rr}$ is the rigid body state vector. $\boldsymbol{M}_{rr}$ is the mass matrix of the system. $\boldsymbol{A}_{rr}$ includes the contributions of flight dynamics, gravity, and tails.

The aerodynamic contributions of the wing now need to be coupled with flight dynamics.

Take the right wing as an example, the right wing aeroelastic dynamics are

$$\dot{\boldsymbol{x}}_w^R = \boldsymbol{A}_{ww}\boldsymbol{x}_w^R + \boldsymbol{B}_{wr}\Delta\boldsymbol{\alpha}_r^R + \boldsymbol{B}_{wg}\boldsymbol{\alpha}_g^R + \boldsymbol{B}_{wf}M_f^R \tag{2}$$

$$\boldsymbol{y}_w^R = \boldsymbol{C}_{ww}\boldsymbol{x}_w^R + \boldsymbol{D}_{wr}\Delta\boldsymbol{\alpha}_r^R + \boldsymbol{D}_{wg}\boldsymbol{\alpha}_g^R + \boldsymbol{D}_{wf}M_f^R \tag{3}$$

where the superscript ^{R} denotes the right wing. $\Delta\boldsymbol{\alpha}_r^R$ denotes the contribution of the rigid body motion to local angle of attack, subtracting he contributions of trim angle of attack. The right wing is excited by three inputs:

- right body angle of attack $\boldsymbol{\alpha}_r^R$. This is a vector, collecting the AOA caused by right body motions on each strip.
- $\Delta\boldsymbol{\alpha}_r^R$ denotes the contribution of the rigid body motion to local angle of attack, subtracting he contributions of trim angle of attack.
- gust angle $\boldsymbol{\alpha}_g^R$. This is also a vector, the gust AOA can be different at various strip locations.
- hinge moment of the right flap M_f^R.

The output vector $\boldsymbol{y}_w^R$ contains the lift, bending moment and torsion moment generated by the right wing.

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The first coupling matrix connects the right body angle of attack $\boldsymbol{\alpha}_r^R$ with rigid body motions, and is the $\boldsymbol{R}_{AOA-rw}$ matrix we learned in the lecture.

$$\Delta\alpha_r^R = \boldsymbol{R}_{AOA-rw}\boldsymbol{x}_{rr} \tag{4}$$

$$\Delta\alpha_r^L = \boldsymbol{R}_{AOA-lw}\boldsymbol{x}_{rr} \tag{5}$$

The second coupling matrix project the total forces and moment generated by the wing to the flight dynamic equation, namely

$$\begin{pmatrix} \mathbf{0} \\ \Delta\boldsymbol{F}_y \\ \mathbf{0} \\ \Delta\boldsymbol{M} \end{pmatrix}_{\text{wing}} = \boldsymbol{R}_{FM-rw}\boldsymbol{y}_w^R + \boldsymbol{R}_{FM-lw}\boldsymbol{y}_w^L \tag{6}$$

Remember the left wing and right wing are symmetrical.

Now, substitute the wing output equation into the flight dynamic equation results in:

$$\begin{aligned}
\boldsymbol{M}_{rr}\dot{\boldsymbol{x}}_{rr} = \quad & \boldsymbol{A}_{rr}\boldsymbol{x}_{rr} + \boldsymbol{B}_{rr}\boldsymbol{\delta} + \boldsymbol{R}_{FM-rw}(\boldsymbol{C}_{ww}\boldsymbol{x}_w^R + \boldsymbol{D}_{wr}\boldsymbol{\alpha}_r^R + \boldsymbol{D}_{wg}\boldsymbol{\alpha}_g^R + \boldsymbol{D}_{wf}M_f^R) && (7)\\
& +\boldsymbol{R}_{FM-lw}(\boldsymbol{C}_{ww}\boldsymbol{x}_w^L + \boldsymbol{D}_{wr}\boldsymbol{\alpha}_r^L + \boldsymbol{D}_{wg}\boldsymbol{\alpha}_g^L + \boldsymbol{D}_{wf}M_f^L) && (8)\\
= \quad & \boldsymbol{A}_{rr}\boldsymbol{x}_{rr} + \boldsymbol{B}_{rr}\boldsymbol{\delta} + \boldsymbol{R}_{FM-rw}(\boldsymbol{C}_{ww}\boldsymbol{x}_w^R + \boldsymbol{D}_{wr}(\boldsymbol{R}_{AOA-rw}\boldsymbol{x}_{rr}) + \boldsymbol{D}_{wg}\boldsymbol{\alpha}_g^R + \boldsymbol{D}_{wf}M_f^R) && (9)\\
& +\boldsymbol{R}_{FM-lw}(\boldsymbol{C}_{ww}\boldsymbol{x}_w^L + \boldsymbol{D}_{wr}(\boldsymbol{R}_{AOA-lw}\boldsymbol{x}_{rr}) + \boldsymbol{D}_{wg}\boldsymbol{\alpha}_g^L + \boldsymbol{D}_{wf}M_f^L) && (10)\\
= \quad & [\boldsymbol{A}_{rr} + \boldsymbol{R}_{FM-rw}\boldsymbol{D}_{wr}\boldsymbol{R}_{AOA-rw} + \boldsymbol{R}_{FM-lw}\boldsymbol{D}_{wr}\boldsymbol{R}_{AOA-lw}]\boldsymbol{x}_{rr} + \boldsymbol{B}_{rr}\boldsymbol{\delta} && (11)\\
& +\boldsymbol{R}_{FM-rw}\boldsymbol{C}_{ww}\boldsymbol{x}_w^R + \boldsymbol{R}_{FM-rw}\boldsymbol{D}_{wg}\boldsymbol{\alpha}_g^R + \boldsymbol{R}_{FM-rw}\boldsymbol{D}_{wf}M_f^R && (12)\\
& +\boldsymbol{R}_{FM-lw}\boldsymbol{C}_{ww}\boldsymbol{x}_w^L + \boldsymbol{R}_{FM-lw}\boldsymbol{D}_{wg}\boldsymbol{\alpha}_g^L + \boldsymbol{R}_{FM-lw}\boldsymbol{D}_{wf}M_f^L && (13)
\end{aligned}$$

The left and right wing dynamics becomes

$$\dot{\boldsymbol{x}}_w^R = \boldsymbol{A}_{ww}\boldsymbol{x}_w^R + \boldsymbol{B}_{wr}\boldsymbol{R}_{AOA-rw}\boldsymbol{x}_{rr} + \boldsymbol{B}_{wg}\boldsymbol{\alpha}_g^R + \boldsymbol{B}_{wf}M_f^R \tag{14}$$

$$\dot{\boldsymbol{x}}_w^L = \boldsymbol{A}_{ww}\boldsymbol{x}_w^L + \boldsymbol{B}_{wr}\boldsymbol{R}_{AOA-lw}\boldsymbol{x}_{rr} + \boldsymbol{B}_{wg}\boldsymbol{\alpha}_g^L + \boldsymbol{B}_{wf}M_f^L \tag{15}$$

Assemble the above equations to the state-space form leads to:

$$\begin{pmatrix} \dot{\boldsymbol{x}}_{rr} \\ \dot{\boldsymbol{x}}_w^R \\ \dot{\boldsymbol{x}}_w^L \end{pmatrix} = \begin{bmatrix} \boldsymbol{A}_{11} & \boldsymbol{A}_{12} & \boldsymbol{A}_{13} \\ \boldsymbol{A}_{21} & \boldsymbol{A}_{ww} & \boldsymbol{0} \\ \boldsymbol{A}_{31} & \boldsymbol{0} & \boldsymbol{A}_{ww} \end{bmatrix} \begin{pmatrix} \boldsymbol{x}_{rr} \\ \boldsymbol{x}_w^R \\ \boldsymbol{x}_w^L \end{pmatrix} + \begin{pmatrix} \boldsymbol{B}_{rg} \\ \boldsymbol{B}_{wg}^R \\ \boldsymbol{B}_{wg}^L \end{pmatrix} \begin{pmatrix} \boldsymbol{\alpha}_g^R \\ \boldsymbol{\alpha}_g^L \end{pmatrix} + \begin{pmatrix} \boldsymbol{B}_\delta \\ \boldsymbol{0} \\ \boldsymbol{0} \end{pmatrix} \begin{pmatrix} \delta_T \\ \delta_e \\ \delta_r \end{pmatrix} + \begin{pmatrix} \boldsymbol{B}_{rf} \\ \boldsymbol{B}_{wf}^R \\ \boldsymbol{B}_{wf}^L \end{pmatrix} \begin{pmatrix} M_f^R \\ M_f^L \end{pmatrix} \tag{16}$$

where

$$\boldsymbol{A}_{11} = \boldsymbol{M}_{rr}^{-1}[\boldsymbol{A}_{rr} + \boldsymbol{R}_{FM-rw}\boldsymbol{D}_{wr}\boldsymbol{R}_{AOA-rw} + \boldsymbol{R}_{FM-lw}\boldsymbol{D}_{wr}\boldsymbol{R}_{AOA-lw}] \tag{17}$$

$$\boldsymbol{A}_{12} = \boldsymbol{M}_{rr}^{-1}\boldsymbol{R}_{FM-rw}\boldsymbol{C}_{ww} \tag{18}$$

$$\boldsymbol{A}_{13} = \boldsymbol{M}_{rr}^{-1}\boldsymbol{R}_{FM-lw}\boldsymbol{C}_{ww} \tag{19}$$

$$\boldsymbol{A}_{21} = \boldsymbol{B}_{wr}\boldsymbol{R}_{AOA-rw} \tag{20}$$

$$\boldsymbol{A}_{31} = \boldsymbol{B}_{wr}\boldsymbol{R}_{AOA-lw} \tag{21}$$

$$\boldsymbol{B}_{rg} = [\boldsymbol{M}_{rr}^{-1}\boldsymbol{R}_{FM-rw}\boldsymbol{D}_{wg}, \boldsymbol{M}_{rr}^{-1}\boldsymbol{R}_{FM-lw}\boldsymbol{D}_{wg}] \tag{22}$$

$$\boldsymbol{B}_{rf} = [\boldsymbol{M}_{rr}^{-1}\boldsymbol{R}_{FM-rw}\boldsymbol{D}_{wf}, \boldsymbol{M}_{rr}^{-1}\boldsymbol{R}_{FM-lw}\boldsymbol{D}_{wf}] \tag{23}$$

$$\boldsymbol{B}_\delta = \boldsymbol{M}_{rr}^{-1}\boldsymbol{B}_{rr} \tag{24}$$

$$\boldsymbol{B}_{wg}^R = [\boldsymbol{B}_{wg}, \boldsymbol{0}] \tag{25}$$

$$\boldsymbol{B}_{wg}^L = [\boldsymbol{0}, \boldsymbol{B}_{wg}] \tag{26}$$

$$\boldsymbol{B}_{wf}^R = [\boldsymbol{B}_{wf}, \boldsymbol{0}] \tag{27}$$

$$\boldsymbol{B}_{wf}^L = [\boldsymbol{0}, \boldsymbol{B}_{wf}] \tag{28}$$

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